
2019-Ph-H1-Q23

Section: Electricity

Topic: Capacitors

Question:

A 100 μF capacitor is charged to 12 V. What is the energy stored?

$$E = \frac{1}{2}CV^2 = \frac{1}{2} \times 100 \times 10^{-6} \times 12^2 = \frac{1}{2} \times 100 \times 144 \times 10^{-6} = 7200 \times 10^{-6} = 7.2 \times 10^{-3} \text{ J}$$

Correct Option: D ($7.2 \times 10^{-3} \text{ J}$)

Guidance for Students:

Watch your units! Capacitance in farads, not microfarads.

Revision Tips:

- Convert μF to F: $1\mu\text{F} = 10^{-6}\text{F}$
 - Use $E = \frac{1}{2}CV^2$
 - Know all three capacitor energy formulas
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